Ada Boost

Given a data set $D = \{(x_i, y_i)\}_{i=1}^{N} \sim \text{iid } \mathbb{P}_{xy}$

$$y_i \in \{-1, 1\}$$

Adaboost Algo:

Init : weights to datapoints uniformly

$$w_i^1 = \frac{1}{N} \qquad i = 1 \dots N$$

For $m = 1$ to M : do

i) Train a weak learner $h_m(x)$, "respecting" the current weights on data w_i^m.

ii) Estimate the weighted error :

$$\epsilon_m = \frac{\sum_{i=1}^{N} w_i^m \mathbb{I}(y_i \neq h_m(x_i))}{\sum_{i=1}^{N} w_i^m}$$

iii) Estimate the optimal step size :

$$\alpha_m = \frac{1}{2} \log\left(\frac{1-\epsilon_m}{\epsilon_m}\right)$$

iv) update the weights on datapoints.

$$w_i^{m+1} = w_i^m \exp\left(-\alpha_m y_i h_m(x_i)\right)$$

v) Find the Ensemble:

$$H_m(x) = H_{m-1}(x) + \alpha_m h_m(x)$$

End For.

AdaBoost as Grad. Boosting

$$H_m(x) = H_{m-1}(x) + \alpha_m h_m(x)$$

$$\alpha_m, h_m = \underset{\alpha, h}{\operatorname{argmin}} \sum_{i=1}^{N} L\left(y_i, H_m(x)\right)$$

For Adaboost, the loss fn. that is used

$$L\left(y, h(x)\right) = \exp\left(-y.h(x)\right) \quad : \text{Exp. loss.}$$

$$\Rightarrow \sum_{i=1}^{N} \exp\left[-y_i\left[H_{m-1}(x_i) + \alpha h(x_i)\right]\right]$$

$$h^*, \alpha^* = \arg\min \sum_{i=1}^{N} \left\{ \exp\left[-y_i H_{m-1}(x_i)\right] \cdot \exp\left[y_i \cdot \alpha \cdot h(x_i)\right] \right\}$$

Consider

$$\sum_{i=1}^{N} \underbrace{\exp\left[-y_i \cdot H_{m-1}(x_i)\right]}_{\text{Ild of } h \,\&\, \alpha} \cdot \exp\left[-y_i \alpha \cdot h(x_i)\right]$$

$$\triangleq w_i^m$$

Note w_i^m is recursive.

i.e., $$w_i^{m+1} = \exp\left[-y_i \cdot H_m(x_i)\right]$$

$$= \exp\left[-y_i \left(H_{m-1}(x) + \alpha_m h_m(x)\right)\right]$$

$$= \exp\left[-y_i H_{m-1}(x)\right] \cdot \exp\left[-\alpha_m y_i h_m(x)\right]$$

$$w_i^{m+1} = w_i^m \cdot \exp\left[-\alpha_m y_i h_m(x)\right]$$

Now, the loss function is,

$$\underset{h,\alpha}{\operatorname{argmin}} \left\{ \sum_{i=1}^{N} w_i^m \cdot \exp\left[-y_i \cdot \alpha \cdot h(x_i)\right] \right\}$$

$$= e^{-\alpha} \sum_{i:\, y_i = h(x_i)} w_i^m + e^{\alpha} \sum_{i:\, y_i \neq h(x_i)} w_i^m$$

The above will be minimized by an $h(x)$ that would minimize the sum of weights on the mis-classified points.

<u>To get optimal Classifier weights (α_m)</u>

Let $$W = \sum_{i=1}^{N} w_i^m$$

by definition, we have

$$\epsilon_m = \frac{1}{W} \sum_{y_i \neq h(x_i)} w_i^m$$

$$= \frac{\sum_{i=1}^{N} w_i^m \mathbb{I}_{y_i \neq h(x_i)}}{\sum w_i^m}$$

$$\therefore \sum_{y_i \neq h(x_i)} W_i^m = W \cdot \epsilon_m$$

$$\sum_{y_i = h(x_i)} W_i^m = W(1 - \epsilon_m)$$

Now consider the loss fn,

$$e^{-\alpha} \sum_{i : y_i = h(x_i)} w_i^m + e^{\alpha} \sum_{i : y_i \neq h(x_i)} w_i^m$$

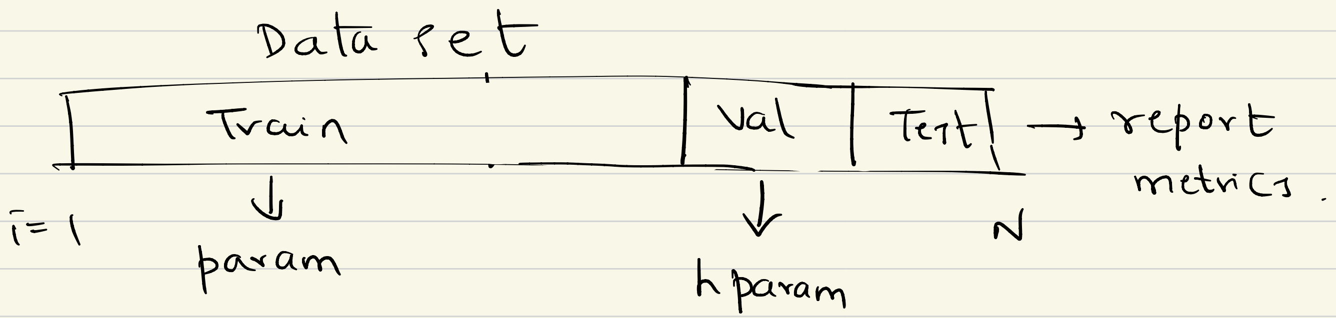
$$L = W \left[e^{-\alpha}(1 - \epsilon_m) + e^{\alpha} \cdot \epsilon_m \right]$$

$$\alpha^* = \underset{\alpha}{\operatorname{argmin}} \; L(\,)$$

$$\frac{\partial L}{\partial \alpha} = 0 \quad ,$$

$$\alpha^* = \frac{1}{2} \log \frac{1 - \epsilon_m}{\epsilon_m}$$

Cross-validation



K-fold cross-validation.

Un-supervised Learning

Estimate the marginals.

Given $D = \{x_i\}_{i=1}^{N} \sim \text{iid } \mathbb{P}_x$

Estimate $\mathbb{P}_x$.

$p_\theta(x)$: model distribution in unsuper. lear.

is taken to be a lat. var. model.

$$p_\theta(x) = \sum_z p_\theta(x, z) \quad \text{or} \quad \int_z p_\theta(x, z)\, dz$$

Once we have a lat. var model.
we can estimate

$$P_\theta(z \mid x)$$

If z is discrete → Clustering

z is continous → feature extraction / representation / Embedding

$$X \xrightarrow{\text{Enc.}} Z \xrightarrow{\text{Dec.}} X$$